

# Osmium

Os (Os I). Strongest classified transitions below 2 μm (NIST): 40 levels, 90 lines.

Os<sup>+</sup> ionisation limit 68058.90 cm<sup>-1</sup> = 8.43823 eV (146.932 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element  $[J||e\mathbf{r}||J']$  in ea<sub>0</sub>  
Matrix elements are measured or NIST values; \* = high-accuracy theory, ≈ = model calculation.  
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation,  $n = 70$ .  
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).

